

Banks' optimal implementation strategies for a risk sensitive regulatory capital rule: a real options and signalling approach *

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Abstract

I evaluate a bank's incentives to implement a risk sensitive regulatory capital rule. The decision making is analyzed within a real options framework where optimal policies are derived in terms of threshold levels of risk. A rating agency's risk evaluation may influence the decision. The rating agency may play an active role and make it optimal for the bank to implement the risk sensitive rule earlier than it otherwise would have preferred. As an example, a bank's decision of whether to use internal rating based models for credit risk (the IRB-approach) under the new Basel accord (Basel II) is evaluated.

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1 Introduction

In order to promote people's trust in the financial sector, authorities often supervise banks and require that they hold a minimum level of capital¹. The aim of this regulatory capital is to provide a cushion that can absorb large and sudden deficits in the bank's earnings and thereby avoid a bank failure. The bank's customers and lenders are also concerned with the bank's riskiness. These outsiders' risk perception may conveniently be represented by a rating agency's judgement of whether the bank is able to meet its obligations during a future time period. The owners of the bank determine the preferred level of capital taking into account both the regulatory capital and the bank's rating. I study the situation where banks are given an option to select between two rules for the computation of regulatory capital. The bank can continue to apply the rule currently in use or switch to the new rule. The decision to switch to the new rule is irreversible, i.e., the bank has to use this rule both today and in the future. The irreversibility is imposed by the authorities because they do not want that banks shift back and forth between different rules. The application of the new rule may give more information to the rating agency about the bank's risk characteristics or may make the rating agency evaluate the bank differently. This may influence the rating of the bank. The bank's decision of whether to implement the new rule will therefore take into account both the changes in regulatory capital and possible adjustments in the rating of the bank. This situation may be viewed as a game between the bank and a rating agency. I analyze the bank's and the rating agency's decision making within a real options framework, where the bank always has an option to delay the

¹I adhere to the convention in the banking literature and use the term capital. For non-banks it is customary to use the term equity.

implementation of the new rule. Implementation is optimal if the state variable representing risk is below a threshold level. The rating agency's role may either be an active or a passive one. When the rating agency has an active role, it may cause the bank to implement the new rule earlier than it otherwise would have preferred to do. This is an example of a situation where the rating agency, or outsiders in general, may have a disciplining effect on bank behavior.

My work adds to the literature covering the application of real options theory in different industries. Textbook treatment of optimal investment timing for irreversible investments may, e.g., be found in Dixit and Pindyck (1994) and Amram and Kulatilaka (1999). My work also contributes to the literature concerning the consequences of a risk sensitive regulatory capital regime. In particular, I describe how banks optimal policies depend on whether they are constrained by the current rule, the reduction in regulatory capital obtained by applying the new rule, and on possible changes in outsider's capital requirement. Different assumptions regarding this capital requirement influences the optimal policies. I also explain how different assumptions leads to different predictions regarding changes in capital for implementing and non-implementing banks. The micro perspective, where I focus on individual banks, complements the macro perspective usually applied when analyzing financial stability issues related to such regimes.

The analysis is inspired by the option given to banks in the Basel accord (Basel II), see Basel Committee on Banking Supervision (2004), where banks can choose between a standard approach and an approach based on internal rating based models (IRB) to compute regulatory capital for credit risk. The latter alternative is more sensitive to changes in risk over time than the first alternative. For completeness I have briefly presented the IRB approach in Appendix B. Within

the Basel II framework, a bank using the standard approach may be seen as using the current rule, here comparable to the non-rating based rule. Note that the potential to use the IRB approach may depend on the type of bank. If statistical methods based on observable variables are required, all major banks, as measured by size, may probably be able to implement a verifiable internal rating systems. This approach may be harder to apply for banks that mainly perform relationship lending. Relationship lending is based on "soft information", see, e.g., Berger and Udell (2002). Soft information is information that is hard to communicate to others (Petersen and Rajan (2002)) or information that cannot be credibly transmitted to others (Stein (2002)). Risk assessments solely based on soft information are rather difficult to quantify and to verify for a regulator. Basel II are based on the so called three pillars. The first pillar covers minimum capital requirement, the second pillar deals with the supervisory review process, and the third pillar covers market discipline. My analysis covers obviously pillar one, but it also deals with a rating agency's, or outsiders', perception of the bank's riskiness. As such, the analysis is also relevant when considering the consequences of market discipline (pillar three) for a bank's selection of regulatory capital (pillar one).

In the analysis the preferred capital ratio is the highest of either regulatory capital or the capital requirement following a rating agency's risk evaluation. The concept of a capital requirement following a rating agency's risk evaluation is here comparable to the concept of a bank's market capital requirement in Berger et al. (1995). They describe on page 395 a bank's market capital requirement as

".. the capital ratio that maximizes the value of the bank in the absence of regulatory capital requirements (and all the regulatory mechanisms

that are used to enforce them), but in the presence of the rest of the regulatory structure that protects the safety and soundness of banks”.

The implication is that the market value of the bank will decline if the the bank has too much or too little capital. The introduction of the risk sensitive rule may influence the bank in two ways. The first effect is *reduced regulatory capital*. A reduction in regulatory capital will reduce the optimal capital ratio, provided that the bank is constrained by the current rule. Because the decision to use the new rule is irreversible, the bank must not only take into consideration the immediate change in optimal capital from the new rule, but also the future development in the difference in regulatory capital between the old and the new rule. The second effect is the *signalling effect*. In short, the signalling effect reflects changes in the outside capital requirement. The capital requirement is changed by the rating agency, possibly upon observing whether the bank switches to the new rule. As an example, under the Basel II rules the internal models are to be approved by regulators. Regulators will only approve models if they are of a sufficient standard. Such an approval may be a signal about the portfolio quality and the quality of the management of the bank. If the bank does not introduce the risk sensitive rule, i.e., does not get the regulator’s approval, the external stake holders may suspect that they in the future will face negative surprises concerning losses on the bank’s loan portfolio. If the bank does not introduce the new rule, the optimal capital ratio may therefore increase.

My basic premise of a unique optimal capital ratio is not trivial from a theoretical perspective. After all, if it was all the same which capital ratio the owners of the bank decided on, the owners would be indifferent when selecting the regulatory

capital and the level of the capital in general. All capital ratios would be optimal. According to the result of Miller and Modigliani (M & M), see Modigliani and Miller (1958), the choice of level of capital does not influence the market value of the company ("the size of the pie"). Any change in the level of capital will only cause a redistribution of value between equity and bond holders (reflecting changing "shares of the pie"). There will be no gain to the shareholders from engaging in the activity of changing the capital ratio. If one makes other assumptions than those in the M & M, there may be an optimal capital ratio. Changing the capital ratio from a non-optimal to an optimal level will then cause the value of the shareholders' holding to increase. This increase may again be caused by an increase in the market value of the company, by a redistribution of wealth from bond holders, or a combination of the two effects. I will now briefly present the main arguments that are put forward in the literature that can explain the existence of an optimal capital ratio. This is not meant as an exhaustive literature review, but rather as a to-the-point presentation of the arguments.

Costs of financial distress make it optimal to avoid holding low levels of capital. Examples of such costs are bankruptcy costs and the costs of foregone business opportunities due to outsiders' unwillingness to conduct business with a company that may fail. Deadweight losses due to bankruptcy and reorganization were mentioned by Modigliani and Miller (1958). *Taxes* favor the use of debt. Interest payments are deductible in the company's taxable income. Increasing the level of debt will therefore reduce the authorities share of profit and leave more to the shareholders, see, e.g., Miller (1976). *Transaction costs* are costs of raising capital. In the presence of transactions cost, the arbitrage argument causing the M & M argument to hold may no longer be strictly valid. Transaction costs also form the

basis for the pecking-order model of debt, see Myers (1984). According to this model, retained earnings are the "cheapest" form of capital, followed by new debt and new equity. The capital ratio will then vary over time with the difference between necessary investment and internally generated funds. Several explanations for an optimal capital ratio are based on the argument of *asymmetric information*. As an example, managers of the bank may use the level of capital as a signal to financial markets about the quality of banks' assets. In Ross (1977) there are two types of companies. One company will have a higher final value than the other. The actual type of a company is not known by the market. If the manager has information about the true type, and with an appropriate incentive structure, the manager will take on relatively more debt in the best type of company in order to maximize his own reward. The market will then price the two types of companies differently. This signal causes an increase in the value of equity for the good company. Another example of asymmetric information is the agency cost argument that increased debt will lead to increased operational efficiency, see Jensen (1986). A requirement to service debt will discipline the managers and induce a more efficient operation of the firm. One argument applying specifically to banks is the presence of a *safety net* for banks' depositors. The safety net refers to the guarantee that authorities give to depositors for the safety of their bank deposits. If the price that banks pay to the authorities for this guarantee is too low relative to the actual risk, there is an incentive for banks to accept too much deposits. For discussions of the capital ratio related to financial institutions in particular, see, e.g. Berger et al. (1995) or Miller (1995). All the reasons mentioned above may, more or less, be present when a given bank is analyzed. An optimal capital ratio may therefore be the result of a trade-off between several factors. Such a mixture

of explanatory factors may therefore be present in the analysis. This was, e.g., the approach taken by Fama and French (2002) when they tested the pecking-order model against what they named a trade-off model of debt.

I now turn to the model before presenting a numerical example in section three. I summarize the main findings in section four.

2 The model

Optimal capital ratio

The bank optimizes its capital under regulatory regime i according to the rule

$$\gamma^{(i)*}(\varepsilon_t, \mu_t, \beta) = \max(\varepsilon_t, \underline{\gamma}^{(i)}(\mu_t, \beta) + b), \quad i \in \{C, N\}, \quad (1)$$

where $\gamma^{(i)*}(\cdot, \cdot, \cdot)$ is the optimal capital ratio, ε_t is the optimal capital ratio in absence of capital regulation, μ_t is the level of credit risk in the bank's loan portfolio, and β is a variable expressing risk characteristics of the bank. The regulatory capital under rule i is $\underline{\gamma}^{(i)}(\cdot, \cdot)$ and b is the constant buffer capital held above the regulatory capital. The bank can either continue to use the current regulatory rule C or the new rule N . If the bank is *constrained* by the regulatory rule, it will according to equation (1) select a capital ratio equal to the regulatory capital with the addition of necessary buffer capital. In this case the bank would be willing to hold less capital in absence of capital regulation ($\varepsilon_t < \underline{\gamma}_t^{(i)} + b$). If the bank is *unconstrained* by the regulatory rule, the optimal capital is equal to the unregulated optimal capital ε_t . The state variable representing expected percentage credit losses on the underlying portfolio over a given future time period, evolves

according to the equation

$$d\mu_t = f(\mu_t, t)dt + \sigma(\mu_t, t)dW_t, \quad (2)$$

where dW_t is the increment of a standard Brownian motion and $f(\cdot, \cdot)$ and $\sigma(\cdot, \cdot)$ may be functions of the state variable and time.

The rating agency² may place the bank in J possible rating categories. The specific rating depends on the characteristics β of the bank and the bank's capital. The lowest level of capital a bank must hold to get a rating j is

$$\underline{\varepsilon}^j = m(j, \beta), \quad j \in J. \quad (3)$$

The bank's type β may, e.g., be defined by whether or not the bank is characterized by a high or low level of risk of large sudden losses (event risk) in the loan portfolio. It is the bank itself that picks the rating category it wants to belong to by adjusting the level of capital. The only effect for the bank from the choice of capital is that a part of the bank's costs is influenced by the capital level. The cost rate for the relevant part of the costs is $g(\gamma_t)$. I assume further that in order to minimize this cost ratio, the bank will always prefer to hold as little capital as possible while maintaining a given rating j , i.e.,

$$\varepsilon_t = \underline{\varepsilon}^j, \quad j \in J. \quad (4)$$

²If the bank is not actually rated by a rating agency, the risk evaluation may be made by other outsiders, such as borrowers, depositors, or bondholders.

Optimal implementation strategy with an exogenously determined rating

Here the unregulated optimal capital ε_t does not depend on the bank's decision of whether to implement the new rule. The value of costs using rule i is³.

$$V_t[K^{(i)}] = E_t^Q \left(\int_t^\infty e^{-r(s-t)} g(\gamma^{(i)*}(\varepsilon_s, \mu_s, \beta)) ds \right), \quad i \in \{C, N\}, \quad (5)$$

where $E_t^Q(\cdot)$ is the expectation operator under the equivalent martingale measure Q conditioned on information at time t and r is the constant instantaneous risk free interest rate.

The payoff from the option to implement the new rule at the exercise date τ equals the value of reduced costs less implementation costs⁴ I_t , i.e.,

$$Z_\tau = V_t[K^{(C)}] - V_t[K^{(N)}] - I_t, \quad \tau = t. \quad (6)$$

The implementation costs may grow with a constant rate δ , i.e., $dI_t = I_t \delta dt$. The market value of the option at time t when exercised at a future time τ , Z_t^τ is then equal to

$$Z_t^\tau = E_t^Q (e^{-r(s-t)} Z_\tau), \quad t \leq \tau, \quad (7)$$

where Z_τ is the option's exercise payoff. With admissible exercise dates $\mathcal{T}$, the

³Equation (5) is the standard way of writing pricing equations in the derivatives and contingent claims literature. I assume that technical conditions hold. Technical conditions may, e.g., be found in Duffie (1992). Another commonly used term for this evaluation approach is *risk neutral* evaluation.

⁴The bank wants to minimize costs. Minimum costs are obtained when the cost reduction is maximized.

market value at time t with the optimal exercise strategy is

$$Z_t^* = \sup_{\tau \in T} Z_t^\tau, \quad t \leq \tau. \quad (8)$$

It is customary to express exercise policies in terms of threshold levels for the state variable μ_t . It is reasonable to assume that policies are such that the exercise of the options are optimal when the risk level μ_t is equal to or lower than a threshold level μ_t^{**} . Waiting, or non-exercise, will then be optimal at higher risk levels. Conditions for securing that the exercise of the option is optimal for $\mu_t \leq \mu_t^{**}$, and waiting to be optimal otherwise, is, e.g., discussed on page 128 in Dixit and Pindyck (1994).

Analyzing the implementation decision by applying equations (5)-(8) is standard in the real options literature. It is well known, see, e.g., McDonald and Siegel (1986), that an option to delay the investment decision may create a positive hurdle that the value of the investment must pass before the investment is made. In our case it is worth noting that it will not be optimal for a bank that is constrained by rule C to implement the new rule if this leads to an immediate increase in regulatory capital.

Proposition 1 (Condition of non-increasing regulatory capital for implementation). *If the implementation cost is zero and if the bank is constrained by regulatory rule C, it will only implement the new rule N if the new rule does not give an immediate increase in regulatory capital.*

Proof. See appendix. □

Large banks where the main investments already have been made may be

considered to have approximately zero implementation costs. For these banks one would not expect to observe an increase in regulatory capital at the implementation time. It is also worth noting that if a bank of a known type is unconstrained by the current rule C, it will not invest in rule N if the investment cost is larger than zero.

Proposition 2 (Non optimality of N if the bank is unconstrained by C). *If the investment cost is larger than zero and if the bank of a known type β^i is always unconstrained by rule C, it will not be optimal for the bank to implement rule N.*

Proof. The result follows from the fact that it is always the rating agency's capital requirement that determines the capital level and thereby the present value of costs. Because the rating agency's capital requirement is not influenced by the choice of regulatory rule (the requirement is exogenously determined), the new regulatory rule may only cause an increase in the capital held by the bank. The value of the cost savings will therefore not be positive and the bank is therefore not willing to pay a positive implementation cost.

□

Optimal implementation strategies with an endogenously determined rating

Here the market's capital requirement may depend on the bank's decision of whether to implement the new rule. This may be considered as a game between the bank, the market and "nature" that is repeated at every point in time. Nature plays a mixing strategy that provides the state variable μ_t . The bank and the

rating agency know the stochastic mechanism determining nature's play and nature's play is independent of the play of the bank and the rating agency. In order to simplify the exposition i therefore in the following restrict the term "player" to the bank and the outsiders.

The strategies at time t is the bank and the outsider's plan for how to play the game at time t , $S_t = \{S_t^{(1)}, S_t^{(2)}\}$. The strategies available to the parties depend on the history h_t of the game. The history at time t is

$$h_t = \{S_s, \mu_s\}, \quad 0 \leq s \leq t, \quad (9)$$

i.e., the history contains the player's previous actions and the history of the state variable up to, and including the current time. The action space at time t , $A(h_t)$ is

$$A(h_t) = \{a_t^{(1)}, a_t^{(2)}\} = \begin{cases} \{\{1_{\{C\}}, 1_{\{N\}}\}, \varepsilon_t\} & \text{if } 1_{\{N\}} \notin h_s, \quad s < t \\ \{\{1_{\{N\}}\}, \varepsilon_t\} & \text{otherwise} \end{cases}, \quad (10)$$

where $1_{\{i\}}$ is an indicator variable equalling one if regulatory rule i is used. The capital requirement⁵ is bounded below by the capital requirement ε^L that is assigned to a β^L bank and above by the capital requirement ε^H that is assigned to a β^H bank, i.e., $\varepsilon^L \leq \varepsilon_t \leq \varepsilon^H$. The bank's information set at time t , $F_t^{(1)}$ consists of the history before the current date and the time t realization of the state variable μ_t , i.e., $F_t^{(1)} = \{h_s, \mu_t\}$, $s < t$. The rating agency's information set contains in addition player 1's action at time t . It is also reasonable to assume that the bank's type is revealed if the new rule is implemented⁶, i.e.,

⁵To simplify the exposition i use the variable ε_t instead of $\underline{\varepsilon}^j$.

⁶I may assume that there is remaining uncertainty regarding the bank's type after the rating agency has observed the bank under the new regulatory rule. Because there is no learning taking

$$F_t^{(2)} = \{h_s, \mu_t, a_t^{(1)}, 1_{\{N\}}\beta\}, \quad s < t, \quad \beta \in \{\beta^L, \beta^H\}.$$

The continuation game starting at time t contains the game played at time t and the games played at all future times. The players' strategies⁷ for this continuation game is $\{S^{(1)} | F_t^{(1)}, S^{(2)} | F_t^{(2)}\}$. Player 1's discounted payoff $U_1(\cdot, \cdot, \cdot)$ from the continuation game is a function of the players' strategies and the bank's type. This payoff corresponds to the costs induced by the capital ratio,

$$U_1(S^{(1)} | F_t^{(1)}, S^{(2)} | F_t^{(2)}, \beta) = -E_t^Q \left(\int_t^\infty e^{-r(s-t)} 1_{\{C\}} g(\gamma^{(C)*}(\varepsilon_t, \mu_t, \beta)) ds \right) \quad (11) \\ - E_t^Q \left(\int_t^\infty e^{-r(s-t)} 1_{\{N\}} g(\gamma^{(N)*}(\varepsilon_t, \mu_t, \beta)) ds \right) - \hat{1}_{\{N\}} I_t,$$

where the indicator variable $\hat{1}_{\{N\}}$ equals one only at the implementation date. The outsiders' beliefs about the bank's type at time t is dependent on the information set at time t , i.e., $p(\cdot | F_t^{(2)})$. The outsider's expected utility at time t is

$$u(S_t^{(1)} | F_t^{(1)}, S_t^{(2)} | F_t^{(2)}, p(\cdot | \cdot)) = -(\varepsilon^L - \varepsilon_t)^2 p(\beta^L | F_t^{(2)}) - (\varepsilon^H - \varepsilon_t)^2 p(\beta^H | F_t^{(2)}) \quad (12)$$

where $u(\cdot)$ is the utility function. The utility function is comparable to a loss function, where the highest obtainable utility is equal to zero. It is optimal to select the highest level of capital if the outsiders believe the bank is of type β^H and the lowest level of capital if it is of type β^L . If the outsiders are uncertain about the bank's type ($p_t(\beta) \neq 1$, $\beta \in \{\beta^L, \beta^H\}$), they will require a level of capital "in between". The first order condition gives the solution $\varepsilon_t = \varepsilon^L p(\beta^L |$

place in the model after the implementation date, this would, however, not add any additional insights.

⁷Note that the players' information sets, and thereby their strategies, are adapted to the filtrated generated by the Brownian motion in equation (2).

$F_t^{(2)}) + \varepsilon^H p(\beta^H | F_t^{(2)})$. Player 2's payoff for the game starting at time t is

$$U_2(S^{(1)} | F_t^{(1)}, S^{(2)} | F_t^{(2)}, p(\cdot | \cdot)) = \mathbb{E} \left(\int_t^\infty e^{-\alpha(s-t)} u(S_t^{(1)} | F_t^{(1)}, S_t^{(2)} | F_t^{(2)}, p(\cdot | F_s^{(2)})) ds \right). \quad (13)$$

where α is a nonnegative discount rate.

With a conditional probability distribution $p(\cdot | \cdot)$ I may then define a Nash equilibrium.

Definition 1. *A Nash equilibrium of the game for the implementation of rule N is described by i)-iii).*

i) *Player 1's optimal policy: For both β^L and β^H ,*

$$U_1(S^{(1)*} | F_0^{(1)}, S^{(2)*} | F_0^{(2)}, \beta) \leq U_1(S^{(1)} | F_0^{(1)}, S^{(2)*} | F_0^{(2)}, \beta), \quad \forall S^{(1)} | F_0^{(1)}.$$

ii) *Player 2's optimal policy:*

$$U_2(S^{(1)*} | F_0^{(1)*}, S^{(2)} | F_0^{(2)}, p(\cdot | \cdot)) \leq U_2(S^{(1)*} | F_0^{(1)}, S^{(2)} | F_0^{(2)}, p(\cdot | \cdot)), \quad \forall S^{(2)} | F_0^{(2)}$$

iii) *Beliefs:*

$$p(\cdot | F_t^{(2)})$$

is well defined for $0 \leq t$.

Solution: Nash equilibrium in the continuation games and Bayesian update of beliefs

I study a particular set of solutions with Bayesian updating of beliefs, wherever possible. Because the bank's type is revealed if it implements the new rule, it is known how the parties will play at and after the implementation date. By default (irreversible decision) the bank will always use the new rule and the rating agency will always require the level of capital corresponding to the bank's type. The continuation games after implementation are therefore all Nash equilibria with no updating of beliefs. Focus must then be on the situation where the bank does not implement. The capital requirement ε_t played by the rating agency in case of no implementation may determine the bank's costs if it does not implement. It will be optimal for the rating agency to get a correct classification of the types of banks as soon as possible. Let the earliest date it is optimal for the bank to implement the new rule dependent on the rating agency's choice of capital ε^i if the bank does not implement be

$$t(\varepsilon^i, \beta) = \min\{s : \mu_s \leq \mu_s^{\varepsilon^i|\beta}, s \geq \hat{t}\}, \quad \varepsilon^L \geq \varepsilon^i \leq \varepsilon^H, \text{ and } \beta \in \{\beta^L, \beta^H\}, \quad (14)$$

where $\mu_s^{\varepsilon^i|\beta}$ is the time s investment threshold for the bank of type β and $\hat{t}$ is the earliest implementation date. The rating agency will require either the highest or lowest level of capital for the non-implementing bank in a situation where only one bank implements. This corresponds to a situation with a separating equilibrium with full revelation of the non-implementing bank's type. With this restriction, following from Bayesian updating, the capital requirement that will result in the

earliest implementation by one of the bank types is

$$\varepsilon^* = \min_{\varepsilon \in \{\varepsilon^L, \varepsilon^H\}} [\mathbf{t}(\varepsilon^H, \beta^L), \mathbf{t}(\varepsilon^L, \beta^H)] , \quad \text{s.t.} \quad \mathbf{t}(\varepsilon^H, \beta^L) \leq \mathbf{t}(\varepsilon^H, \beta^H) \quad \text{and} \quad (15)$$

$$\mathbf{t}(\varepsilon^L, \beta^H) \leq \mathbf{t}(\varepsilon^L, \beta^L),$$

where the restrictions secure that the non-implementing bank receives an correct capital requirement. The type of bank that will potentially implement at the earliest time is β^* , where

$$\beta^* = \begin{cases} \beta^L & \text{if } \varepsilon^* = \varepsilon^H \\ \beta^H & \text{if } \varepsilon^* = \varepsilon^L \end{cases} . \quad (16)$$

The earliest time the investment will be made is therefore $\mathbf{t}(\varepsilon^*, \beta^*)$. I am now ready to characterize the equilibrium corresponding to earliest possible implementation, possibly induced by the rating agency.

Proposition 3 (Equilibrium for the implementation game). *Let the variable β^* be given by (16), the variable $\mathbf{q}$ be the ex-ante probability that the bank is of type β^H , $0 < \mathbf{q} < 1$, and let ε^q be the outside capital requirement based on $\mathbf{q}$. If ε^* in (15) is either ε^H or ε^L , then i)-iii) is a Nash equilibrium of the implementation game with Bayesian updating of beliefs wherever possible.*

i) *Banks' optimal conditional exercise policy:*

$$\text{For } \beta \in \{\beta^L, \beta^H\}, \quad S^{(1)*} \mid F_0^{(1)} = \begin{cases} S_t^{(1)} = 1_{\{C\}} \text{ for } t < \mathbf{t}(\varepsilon^*, \beta) \\ S_t^{(1)} = 1_{\{N\}} \text{ for } t \geq \mathbf{t}(\varepsilon^*, \beta) \end{cases}$$

ii) *The rating agency's optimal conditional capital requirement:*

$$S^{(2)*} | F_0^{(2)} = \begin{cases} S_t^{(2)} = \varepsilon^q & \text{if } t < t(\varepsilon^*, \beta^*) \\ S_t^{(2)} = \varepsilon^* & \text{if } t \geq t(\varepsilon^*, \beta^*) \text{ and } 1_{\{N\}} \notin F_t^{(2)} \\ S_t^{(2)} = \varepsilon^L & \text{if } t \geq t(\varepsilon^*, \beta^*), 1_{\{N\}} \in F_t^{(2)}, \text{ and } \beta = \beta^L \\ S_t^{(2)} = \varepsilon^H & \text{if } t \geq t(\varepsilon^*, \beta^*), 1_{\{N\}} \in F_t^{(2)}, \text{ and } \beta = \beta^H \end{cases}$$

iii) *Beliefs: Conditional probabilities: $\gamma_t^{(N)}$:*

$$p(\beta^* | F_t^{(2)}) = \begin{cases} \emptyset & \text{if } t < t(\varepsilon^*, \beta^*) \\ 0 & \text{if } t \geq t(\varepsilon^*, \beta^*) \text{ and } 1_{\{N\}} \notin F_t^{(2)} \\ 0 & \text{if } t \geq t(\varepsilon^*, \beta^*), 1_{\{N\}} \in F_t^{(2)}, \text{ and } \beta \neq \beta^* \\ 1 & \text{if } t \geq t(\varepsilon^*, \beta^*), 1_{\{N\}} \in F_t^{(2)}, \text{ and } \beta = \beta^* \end{cases}$$

Proof. See the appendix. □

The intuition of the equilibrium strategies in the implementation game is depicted in Figure 1. Note that Proposition 3 does not apply to the case where ε^* is equal to both ε^L and ε^H . In this case the rating agency will not be certain about the non-implementing bank's type.

We may consider two roles for the rating agency in this game. With what may be termed a *passive role* for the rating agency, the rating agency cannot influence the bank's decisions by its non-implementation play. An example of this is the case where the bank is unconstrained by rule C even if the rating agency require the highest level of capital ε^H . This because the optimal capital ratio in equation (1)

under C is not influenced by the rating agency's actions. In this case the investment thresholds and thereby the earliest date for implementation, i.e., $t(\varepsilon^i, \beta)$, is the same for all ε^i and for both types of banks. There may, however, also be a situation with an *active role* for the rating agency. This is the case when the rating agency's strategy may change the bank's optimal capital ratio under rule C according to equation (1). A particular case where the active role played by the rating agency is important, is when the β^H bank never is constrained by regulatory rule C if its type is known with certainty. This β^H bank will not implement the new rule, according to Proposition 2. The rating agency's no-implementation play will here raise the capital level for both types of banks and create a strong incentive for the β^L bank to implement. The β^L bank is the only bank that will implement the new rule.

3 Internal models for credit risk under Basel II

- numerical computation of threshold levels

Internal models and credit risk

Basel II allows for the use of internal models to measure credit risk and to use the measured risk when computing regulatory capital. A short summary of the IRB approach under Basel II is given in Appendix B. The rule requires that the probability of default (PD) and loss given default (LGD) are specified in order to compute regulatory capital for a specific loan under the IRB approach. In order to simplify the exposition I base my computation only on expected percentage credit losses μ_t on an underlying portfolio. I extract the average probability of default

at time t from expected losses by dividing by the parameter LGD, i.e.,

$$PD_t = \max [0, \min [1, \mu_t/LGD]] , \ 0 < LGD \leq 1 . \quad (17)$$

I consider the case where the instantaneous change in expected percentage credit losses on the portfolio develops according to an Ornstein-Uhlenbeck process,

$$d\mu_t = \kappa(\theta - \mu_t)dt + \sigma dW_t, \quad (18)$$

where θ , κ , and σ are nonnegative constants and where dW_t is the increment of a standard Brownian motion. The parameter θ is the long term mean of expected losses. The speed of reversion to the long run mean is captured by the parameter κ . The process (18) corrected for the price of risk is

$$d\mu_t = \kappa(\theta - \frac{\lambda\sigma}{\kappa} - \mu_t)dt + \sigma dW_t^* , \quad (19)$$

where W_t^* is a Brownian motion under an equivalent martingale measure, and λ is the price of risk related to unexpected changes in expected losses. The price of risk equals the required compensation beyond the risk free interest rate $(\eta - r)$ per unit of risk σ , i.e.,

$$\lambda = \frac{\eta - r}{\sigma}. \quad (20)$$

One procedure to determine the price of risk is to compute the required expected return (η) on holding an asset influenced by the specific risk (σ) by applying the CAPM, see, e.g., p. 115 in Dixit and Pindyck (1994) or Nordal (2001) for an application. I would expect the price of risk in (20) to be negative because it is

likely that expected losses (losses are measured as a positive number) are negatively correlated with the return on the market portfolio. I may alternatively write (19) as

$$d\mu_t = \kappa(\theta^* - \mu_t)dt + \sigma dW_t^*, \quad (21)$$

where $\theta^* = (\theta - \lambda\sigma/\kappa)$, see, e.g., Bjerk Sund and Ekern (1995) or Schwartz (1997). The effect of the correction for the price of risk is therefore to reduce (increase) the "long term mean" if the price of risk is positive (negative).

Value and threshold levels

I build a trinomial tree for the state variable μ_t as in Hull and White (1994). Table 1 shows the assumptions for the benchmark example. In the benchmark example I do not differ between types of banks. The unregulated optimal capital is a constant ε equal to 6 per cent. I have used a risk free interest rate of 4 per cent, a price λ per unit of risk in the development in expected losses of -1 per cent, and a volatility of expected losses σ equal to 0.5 per cent. I have further used a time step of 0.25 (quarters). The long run mean of expected losses in the risky portfolio is 0.5 per cent. The long run mean under the risk neutral probability is 0.51 per cent. The value of the new rule is found by using an evaluation period of 50 years. For every node in the tree the present value of the advantage of using the new rule instead of the old during the next time period Δt is computed as⁸

$$V_t \left(\int_t^{t+\Delta t} e^{-r(s-t)} (\gamma^{(C)*}(\mu_s) - \gamma^{(N)*}(\mu_s)) ds \right) \approx \frac{(1 - e^{-r\Delta t})}{r} (\gamma^{(C)*}(\mu_t) - \gamma^{(N)*}(\mu_t)). \quad (22)$$

⁸This means that the cost rate $g(\gamma_t)$ is equal to $1 \times \gamma_t$.

The regulatory capital is 8.0 per cent for loans under the current rule C. The regulatory capital under the new rule is derived by using the IRB model in Basel II with and without a reduction for small and medium size entities (SMEs), see Appendix B. Figure 2 shows the regulatory capital for different levels of expected losses. The break even level of expected losses μ_t making the regulatory capital with rule C equal to the regulatory capital with rule N with no reduction for SMEs is approximately 0.6 per cent. This corresponds to a PD of approximately 1.3 per cent.

(insert Table 1 approximately here)

(insert Figure 2 approximately here)

Figure 3 shows the value of the "immediate implementation" and the "option to implement" alternatives for different levels of current expected losses μ_t . The values decrease when current expected losses increase. I have shown the values for different levels of the parameter representing the force of mean reversion κ . The curves showing the values become less sensitive to the current level of risk when the half life is reduced (i.e., when the force of mean reversion increases). With a strong force of mean reversion, the current risk level means less for the value. The implementation thresholds are approximately 0.9 and 0.6 per cent when the half lives are, respectively, 1 and 5 years. Table 2 shows the implementation thresholds with and without a "wait" option for different assumptions about SME compensation, the force of mean reversion, the size of the implementation cost I_0 , and the rate of reduction δ in implementation costs over time. The investment threshold is much lower when the bank has an option to wait (μ^{**}) compared to situation where it has no such option (μ^*). Future reductions in implementation

costs reduces the implementation thresholds further. Note that this reduction is not included in the evaluation of the "now or never" implementation alternative. The investment thresholds are generally higher with a full SME compensation. This reflects that the value of the option to use the new rule is higher with full SME compensation.

Figure 4 shows the implementation threshold for a type β^L bank that faces a non-implementation capital requirement of ε^H . The bank will be assigned an outside capital requirement of 6 per cent when its type is known. The implementation threshold does not increase and is approximately 0.7 per cent as long as the bank is constrained by rule C. In this case there is no active role for the rating agency. There is an active role for the rating agency only when the bank becomes unconstrained by the current rule. This happens when the outside capital requirement is higher than 10 per cent. If the capital requirement is 11 per cent, the investment threshold is approximately 1.3 per cent.

(insert Figure 3 approximately here)

(insert Table 2 approximately here)

(insert Figure 4 approximately here)

4 Summary

I have analyzed banks' incentives to introduce a risk sensitive rule for regulatory capital. The underlying premise is that banks prefer to have as little capital

as possible within the constraint of regulatory capital and a capital requirement following from a rating agency's risk evaluation. I first study the case where the rating agency's risk evaluation is exogenously determined. I then study a game where the rating agency may update its belief about the type of bank in the case that the bank does not implement. In general, banks' optimal policies will depend on whether they are constrained by the current rule, the reduction in regulatory capital obtained by applying the new rule, and on possible changes in the capital requirement following from changes in the rating agency's risk evaluation. I explain how different assumptions regarding this capital requirement influences the optimal policies. I also explain how different assumptions leads to different predictions regarding changes in capital for implementing and non-implementing banks. I present a numerical example for the use of internal models for measuring credit risk under the Basel II approach. The evaluation framework may more generally be used to evaluate banks' decision making in situations where banks are given an option to irreversibly select between a set of regulatory rules.

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A Proofs

Proposition 1

Proof. I follow the approach on page 128 in Dixit and Pindyck (1994). The optimal policy is derived from the Bellman equation

$$\Omega(\mu_t) = \max \left[B(\mu_t), E_t^Q (\Omega(\mu_{t+dt})) e^{-r dt} \right], \quad (23)$$

where the value function $\Omega(\mu_t)$ is equal to the market value of the implementation option. The present value of the cost reduction if rule **N** is implemented at time t is $B(\mu_t)$. If the rule is not implemented, the bank gets the present value of the option value at time $t + dt$. Exercise is optimal if the exercise value is larger than the continuation value. Note that

$$B(\mu_t) = g(\gamma^{(C)*}(\mu_t)dt - g(\gamma^{(N)*}(\mu_t)dt + E_t^Q (B(\mu_{t+dt})) e^{-r dt} \quad (24)$$

We may then rewrite (23) as

$$\begin{aligned} \Omega(\mu_t) - B(\mu_t) = & \max[0, -g(\gamma^{(C)*}(\mu_t)dt + g(\gamma^{(N)*}(\mu_t)dt + \\ & E_t^Q (\Omega(\mu_{t+dt}) - B(\mu_{t+dt})) e^{-r dt}]. \end{aligned} \quad (25)$$

The left hand side of (25) is a nonnegative number and continuation is only optimal if the continuation value is larger than zero. Because

$$E_t^Q (\Omega(\mu_{t+dt}) - B(\mu_{t+dt})) e^{-r dt} \quad (26)$$

is nonnegative, the optimal capital ratio with rule N must be smaller than with rule C if stopping (exercise of the option) is optimal, i.e., if $-g(\gamma^{(C)*}(\mu_t)) + g(\gamma^{(N)*}(\mu_t)) \leq 0$. Because of the assumption that the bank is constrained by rule C, it is only an increase in regulatory capital that may cause an increase in the optimal capital ratio if rule N is implemented.

□

Proposition 3

Proof. Two cases must be considered. In the first case it is optimal for the rating agency to set the highest capital requirement ε^H at the earliest implementation date, i.e., $\varepsilon^* = \varepsilon^H$. This means that the first bank that will implement in a potential separating equilibrium is the β^L bank, i.e., $\beta^* = \beta^L$. Conditional on receiving a capital requirement of ε^H if it does not invest, it is optimal for both banks to invest the first time that s that $\mu_s \leq \mu_s^{\varepsilon^H|\beta}$, $\hat{t} \leq s$. The rating agency's optimal policy is determined by the choice of minimum capital level ε_t and beliefs $p(\cdot | \cdot)$. If the bank implements, the bank's type is known and the rating agency chooses the optimal capital level. In the time period where none of the banks would implement (before time $t(\varepsilon^H, \beta^L)$), no implementation is observed and the probability of the type of bank is not updated. Implementation would here be a zero probability event. If it is optimal for the β^L bank to implement before the β^H bank, the probability that the non-implementing bank is the β^H bank is equal to one. This probability formation is based on a Bayesian update. Let $t' = t(\varepsilon^H, \beta^L)$ and $t'' = t(\varepsilon^H, \beta^H)$, $A = p(\beta^H | S_t^{(1)} = 1_{\{C\}} \in F_t^{(2)}, t = t' < t'')$ and $B = p(\beta^L | S_t^{(1)} = 1_{\{C\}} \in F_t^{(2)}, t = t' < t'')$. The Bayesian updated probability that

the implementing bank is the β^H bank is then

$$\begin{aligned} p(\beta^H | S_t^{(1)} = 1_{\{C\}} \in F_t^{(2)}, t = t' < t'') &= \frac{p(\beta^H) \times A}{p(\beta^L) \times B + p(\beta^H) \times A} \\ &= \frac{(q \times 1)}{(1 - q) \times 0 + q \times 1} = 1. \end{aligned}$$

If both types of banks would implement the new rule at the first possible implementation date $\hat{t}$, the non-implementation play for the rating agency is a zero probability event according to the Bayesian updating formula. In this case the conditional probability of a β^H bank conditioned on no implementation is set equal to one by assumption.

In the second case it is optimal for the rating agency to set the lowest capital requirement ε^L at the earliest implementation date, i.e., $\varepsilon^* = \varepsilon^L$. The first bank that will implement in a potential separating equilibrium is in this case the β^L bank, i.e., $\beta^* = \beta^L$. The rest of the proof then proceeds as for the first case.

□

B Capital adequacy regulation

The weighted capital ratio at time t , γ_t , is given by

$$\gamma_t = \frac{S_t}{\sum_i w_t^i A_t^{(i)}}, \quad (27)$$

where S_t is capital, $A_t^{(i)}$ is the loan in category i , and w_t^i is a weight for loan in asset category i . Under the Basel II rules, see Basel Committee on Banking Supervision (2004) a bank can choose between three alternatives regarding credit risk; the

standardized approach, the Internal rating based (IRB) foundation approach, and the IRB advanced approach. Under the standardized approach the weights are constants. Under the IRB approaches the weights are computed based on estimates of probabilities of default (PD) and expected loss given default (LGD). With the IRB foundation approach it is assumed that the loss given default is 45 per cent of the exposure. Under the IRB advanced approach, the bank also uses own estimates for LGD. The IRB approach can be reconciled with value-at-risk models for credit risk, see Gordy (2003).

The procedure to compute the necessary capital under the IRB approach involves several steps. First the required capital S per unit of currency (corresponding to the risk weight per one unit of a loan) is computed according to the formula

$$S = [\text{LGD } N(d) - \text{LGD } \text{PD}] \frac{1 + (M - 2.5)b}{1 - 1.5b}, \quad (28)$$

where

$$d = \left(\frac{1}{1 - R} \right)^{0.5} N^{-1}(\text{PD}) + \left(\frac{R}{1 - R} \right)^{0.5} N^{-1}(0.999), \quad (29)$$

M is the effective maturity, and $N(\cdot)$ is the cumulative normal standard distribution. The maturity adjustment factor b is

$$b = (0.11852 - 0.05478 \ln(\text{PD}))^2. \quad (30)$$

The correlation factor R is⁹

$$R = 0.12 \left(\frac{1 - e^{-50PD}}{1 - e^{-50}} \right) + 0.24 \left(1 - \frac{1 - e^{-50PD}}{1 - e^{-50}} \right). \quad (31)$$

The correlation factor R^{SME} for small- and medium-sized entities (SME) is given by the formula

$$R^{SME} = R - 0.04 \left(1 - \frac{s - 5}{45} \right), \quad 5 \leq s \leq 50, \quad (32)$$

where R is given by (31) and s is total annual sales.

Risk weighted assets RWA are computed according to the formula

$$RWA = S12.5EAD, \quad (33)$$

where EAD is exposure at default measured in units of currency.

⁹For residential mortgage exposures $R = 0.15$ and

$$S = [LGD \ N(d) - LGD \ PD)],$$

where d is given by (29).

Tables

Table 1: Parameter values used in the numerical example

Parameter	Numerical value	Description
Δt	0.25	time step
r	0.04	risk free interest rate
λ	-0.01	price of risk
T_N	10	length of option to implement in years
T	50	evaluation period in years
I	0.10	investment expenditures
δ	0.00	depreciation rate for investment expenditures
θ	0.005	long run mean in percentage losses
κ	0.23	speed of mean reversion
σ	0.005	volatility of percentage losses
LGD	0.45	loss given default
b	0.02	buffer capital
ε	0.06	unregulated optimal capital

Table 2: Implementation thresholds with and without a delay option

I_0	δ	SME = 0				SME = 1			
		Half life = 3		Half life = 5		Half life = 3		Half life = 5	
		μ^*	μ^{**}	μ^*	μ^{**}	μ^*	μ^{**}	μ^*	μ^{**}
10	0.0	1.0	0.1	0.7	-0.1	6.8	0.7	4.1	0.6
	-1.0	1.0	-0.1	0.7	-0.2	6.8	0.5	4.1	0.4
	-2.0	1.0	-0.2	0.7	-0.2	6.8	0.4	4.1	0.3
	-3.0	1.0	-0.2	0.7	-0.3	6.8	0.3	4.1	0.2
5	0.0	1.4	0.1	1.0	0.0	7.9	0.9	4.8	0.7
	-1.0	1.4	0.1	1.0	0.0	7.9	0.7	4.8	0.6
	-2.0	1.4	0	1.0	-0.1	7.9	0.6	4.8	0.5
	-3.0	1.4	0	1.0	-0.1	7.9	0.6	4.8	0.5

Numbers are in per cent. Threshold levels μ^* without an option to wait are found by setting the implementation value equal to zero. Threshold levels μ^{**} with an option to wait are found by finding the highest value of expected losses making the value of the "implement now" alternative equal to the value of the option to implement. In both cases the threshold levels are approximate. SME = 0 means that there is no adjustment for small and medium size enterprises. SME = 1 means that there is full adjustment for small and medium size enterprises.

Figures

Figure 1: The implementation decision

The bank of type β ($\beta \in \{\beta^L, \beta^H\}$) will implement the new rule provided that the state variable is lower or equal to the threshold $\mu^{\varepsilon^H, \beta}$ at time t , $t \geq \hat{t}$. The figure shows three paths of the state variable μ_t , each with a different starting point. In the situation described by path A, neither type of bank will invest at time $\hat{t}$. At time t' banks of type β^L will invest. The market will then know that the banks that did not invest are of type β^H , and the capital requirement for such banks is assigned to them. At time t''' , the banks of type β^H will also implement the new rule. In the situation described by path B, banks of type β^L will immediately implement at time $\hat{t}_N$. The non-implementing banks belongs to type β^H . For path C, both types of banks will implement at the earliest possible time.

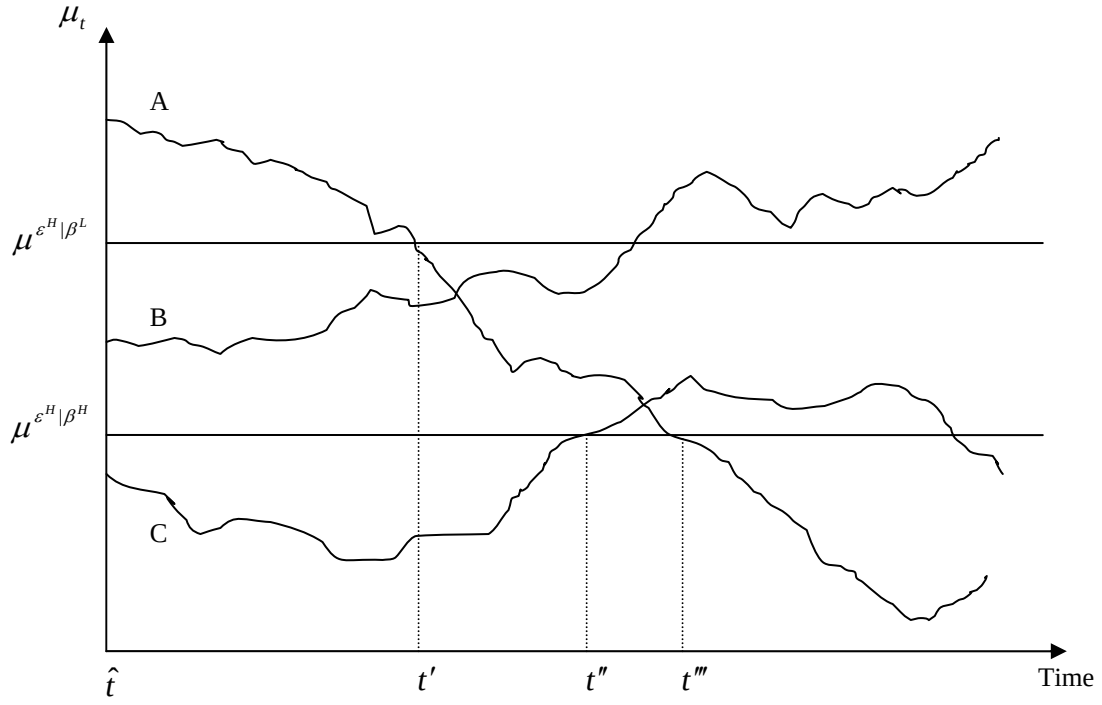


Figure 2: Current and new regulatory rule

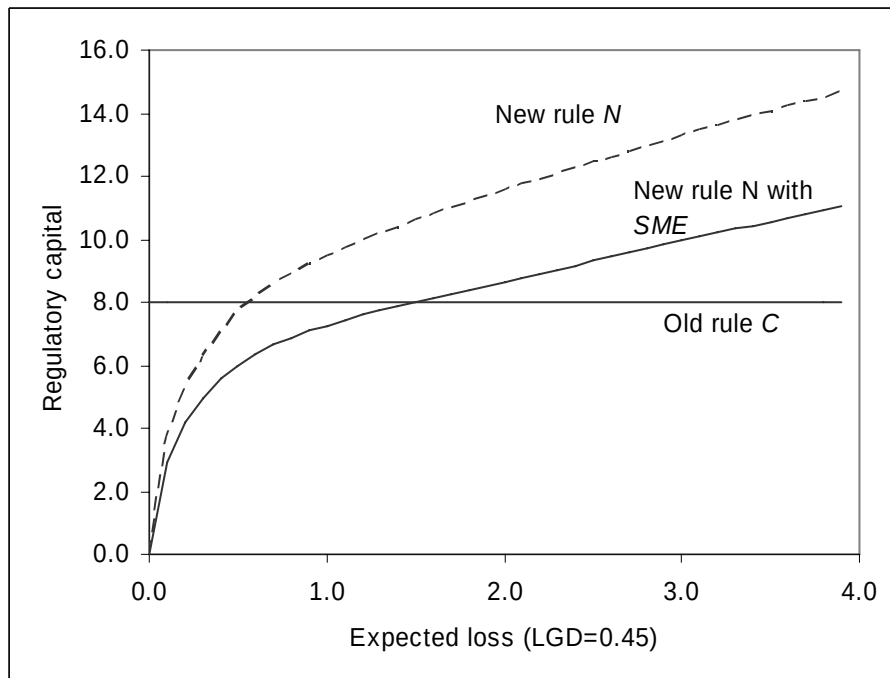


Figure 3: Exercise and option values for the benchmark example

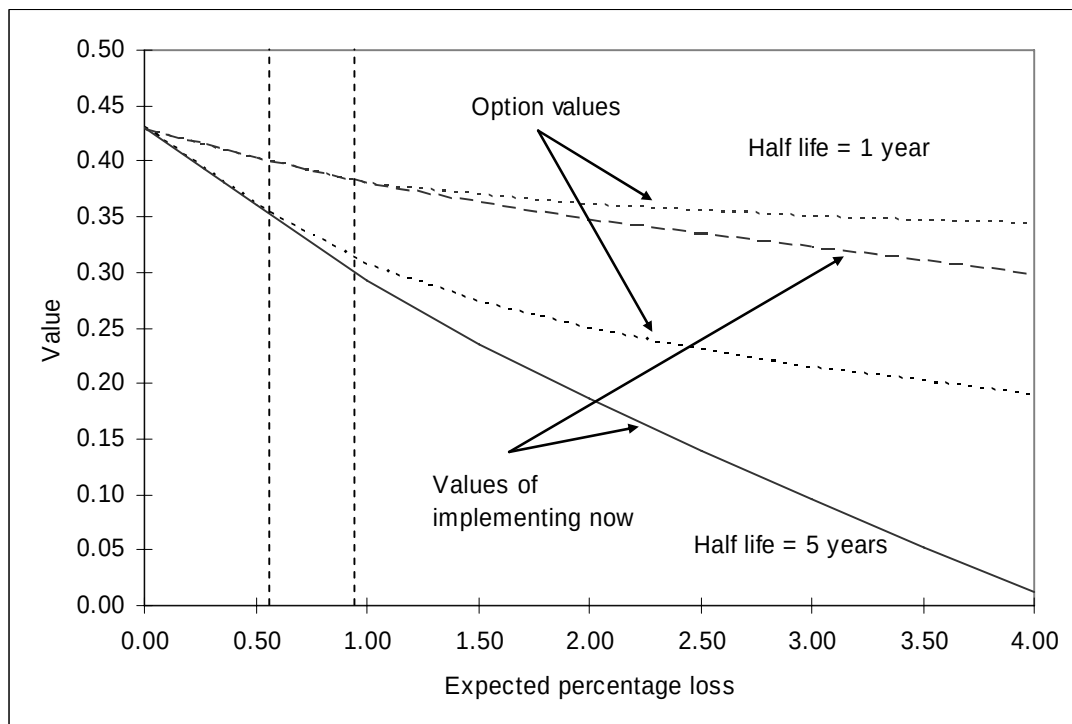


Figure 4: Implementation threshold as a function of non-implementation capital requirement

